



AceBoard - an Arduino-Based Digital Tennis Scoreboard

Transforming tennis scorekeeping with a portable, affordable digital scoreboard designed for schools and community courts—so players can focus on the next point, not the scoreboard.

An independent project by Neha Ashok

About This Project

A decade of competitive tennis, from USTA Youth tennis to varsity first singles, revealed a recurring problem that unexpectedly led me down an engineering path.

Existing scoreboards - the manual flip-charts - consistently disrupted focus and mental game during crucial matches. This wasn't just an inconvenience; it was a barrier to peak performance and enjoyment.

Combining my athlete's perspective with my background in VEX robotics, I recognized an opportunity to apply embedded systems engineering to create a seamless, intuitive scoring experience.

This project is more than just circuits and code; it's a testament to my belief that impactful engineering stems from personal experiences. By understanding the nuanced needs of tennis players, I aimed to genuinely enhance a real-world setting, leveraging technology for improvement, not just novelty.



Why This Project Matters

The Scorekeeping Gap

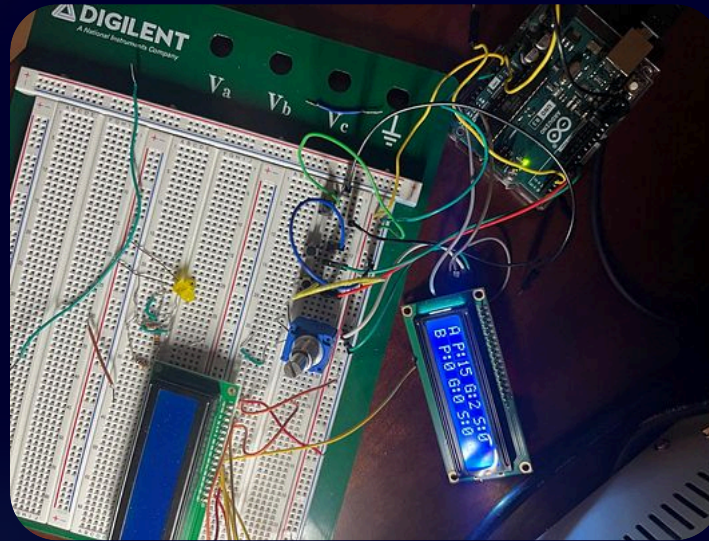
Many school and community tennis courts lack reliable scoreboards. Players mostly rely on memory during matches, leading to confusion, disagreements, and disrupted play—especially during long, high-pressure games. During matches sanctioned by the USTA and high school competition, manual flip-chart scoreboards often disrupt match flow. Players have to pause, physically update scores, resulting in a break in concentration during critical moments. Existing digital scoreboards are expensive, permanently installed, or require infrastructure unavailable at ordinary courts.

This gap is particularly evident in high school tennis, where matches are frequent but resources are limited. The need exists for an affordable, portable solution designed for real tennis courts rather than professional venues.

Personal Connection

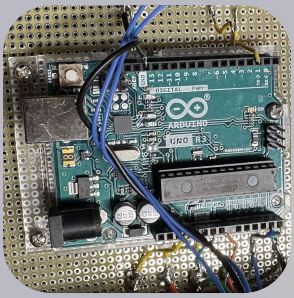
As a first singles player with 10+ years of competitive experience and VEX robotics background, I've witnessed how scoreboard absence disrupts focus and momentum. This project combines my athletic experience with engineering skills to solve a problem I encounter every time I step on court.

Early Prototype: Proof of Concept



The early prototype prioritizes functionality over appearance—a setup using a solderless breadboard with exposed wiring focused on validating the concept before incorporating the design into a ruggedized setup. This stage confirms that the hardware and firmware aspects of the scoring system work reliably.

Prototype Components



Microcontroller & Logic

The input from the user remote is recognized by the IR receiver and mapped to a digital input on the Arduino microcontroller. A signal debounce of 300ms has been incorporated in the logic to prevent accidental button presses from incrementing the scoreboard. Logic incorporated to update scoreboard based on game, set and tiebreak rules.



Display System

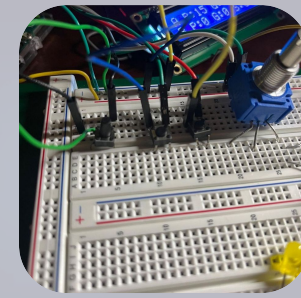
Display is an LCD screen which uses an I2C communication protocol with the Arduino microcontroller. It provides a current score including the game, set and tiebreak status of each player independently.



Input Controls

Push buttons evolved to IR remote for wireless interaction. Explores remote score control for umpires or players on court.

Score updates via button inputs. OK button increments the score, * decrements the score in case of erroneous click of OK button, # button resets the entire scoreboard



Breadboard Wiring

Jumper wires were used for connecting the IR transmitter/receiver to the Arduino inputs and also to wire the I2C connections from the Arduino to the LCD display. The setup uses a solderless breadboard to handle the power and ground connections.

Prototype Challenges

Mounting and Display Challenges

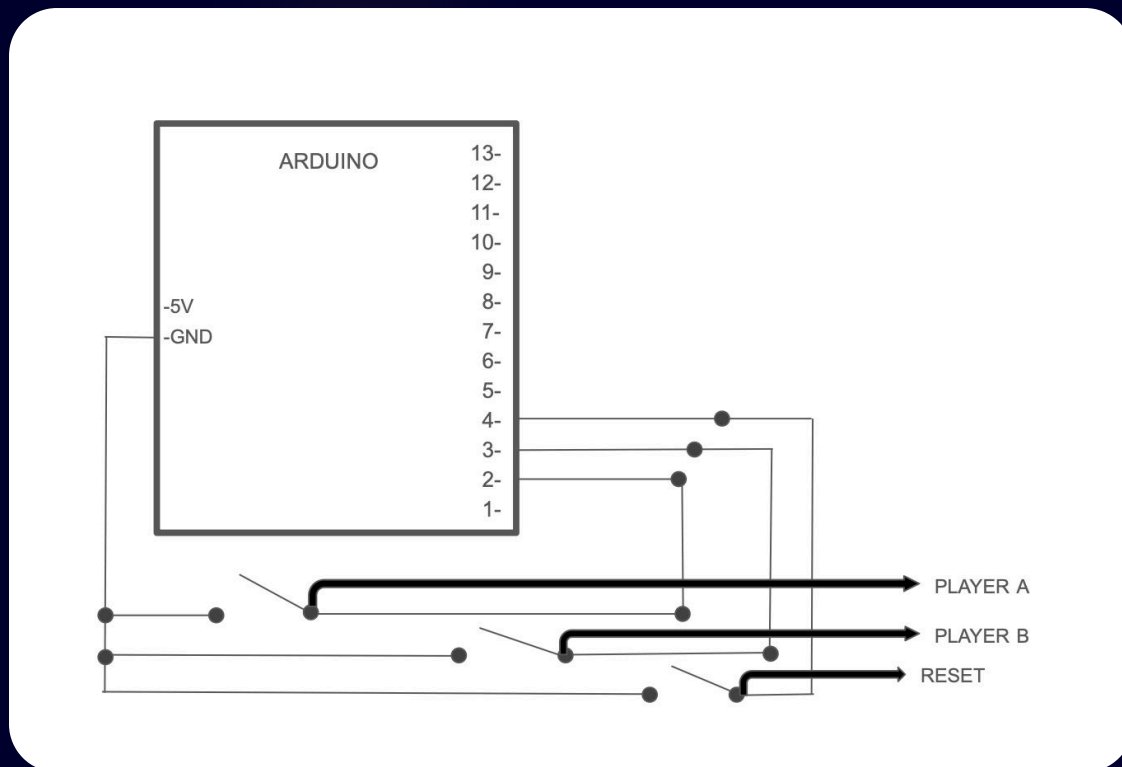
IR receivers sometimes are not in the line of sight of the transmitter. The circuit also does not have an ON/OFF Switch to save battery life. Display is connected to the breadboard through wires.

Small LCDs unsuitable for long distance visibility. Future improvements could incorporate larger LED displays with higher brightness for better readability. This requires a larger power source which is not feasible with the current 9V battery based setup. This would require a large battery/inverter setup or solar power (for outdoor courts).

Wire Management

Breadboards and jumper wires lack mechanical security, are vulnerable to weather, and cannot be used on active courts. Electronics must be enclosed and protected.

Wiring Diagrams

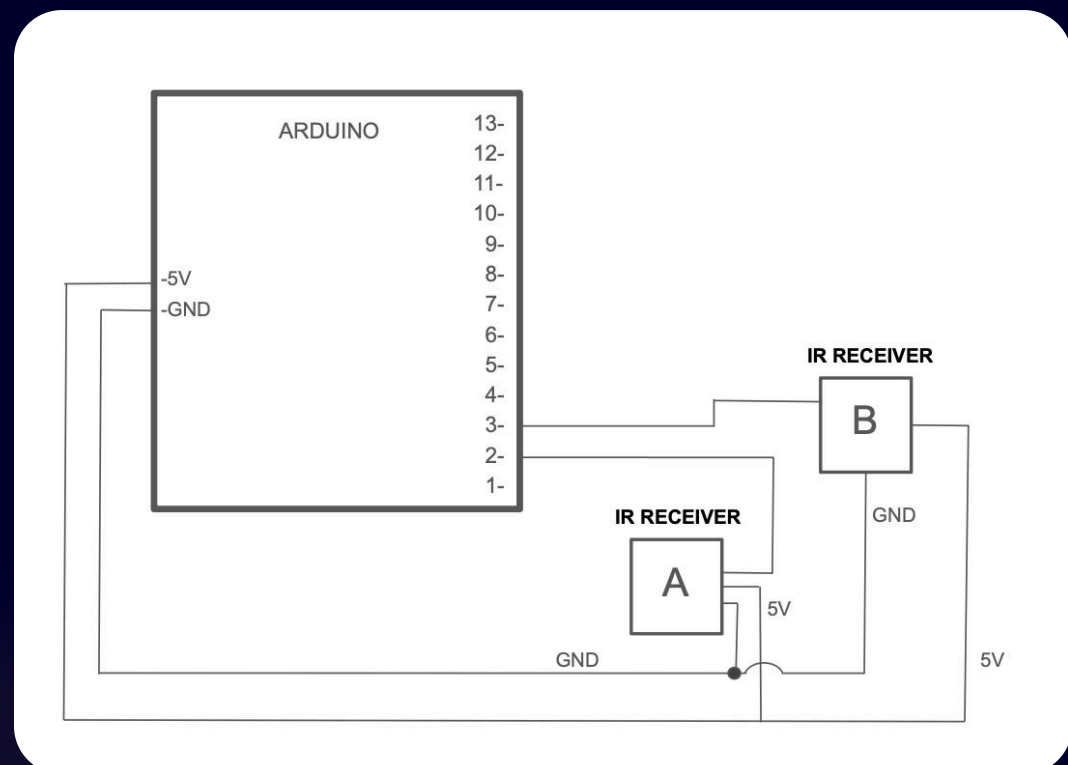


Push Button Configuration

Arduino wired to 3 external push button switches for Player A, Player B, and reset. Internal pull-up provides current pathway when switch is open. Pressing increments game score, set score, point score, or tie-break score depending on game status. 300ms debounce prevents accidental presses.

Infrared Remote System

Logic modified to replace push buttons with two 433 MHz infrared transmitter/receiver kits. Remote controls allow either player to increment scoreboard. Receivers programmed to Arduino pins 2 and 3. OK button increments, * resets game, # resets scoreboard, 1 decrements by a point. Range tested simultaneously to 40 feet. Future improvements include larger LCD via WiFi interface.



User Input & Control System



Button Input Handling

Physical buttons increment scores for each player and reset matches. Program reads button presses in main loop and updates match state accordingly. Simple approach remains reliable during fast-paced matches.

```
if (digitalRead(btnA) == LOW) {  
    pointsA++;  
    checkGameWin();  
    checkSetWin();  
    updateDisplay();  
}
```

Button Press

Player or umpire presses physical button or IR remote

1

Win Check

Logic evaluates game, set, or match completion

3

State Update

Arduino increments appropriate score variable

2

Display Refresh

LCD updates to show current match state

4

Scoring Logic & Tennis Rules Implementation

1

Match State Variables

Separate variables track points, games, sets, and tiebreaks, mirroring actual tennis scoring. Allows independent evaluation of each match component.

```
int pointsA = 0, pointsB = 0;
int gamesA = 0, gamesB = 0;
int setsA = 0, setsB = 0;
bool inTiebreak = false;
int tbA = 0, tbB = 0;
```

2

Point Conversion

Helper function converts internal values to tennis scoring terms (0, 15, 30, 40). Keeps internal logic simple while displaying proper tennis format.

```
String pointText(int p) {
  if (p == 0) return "0";
  if (p == 1) return "15";
  if (p == 2) return "30";
  if (p == 3) return "40";
  return "";
}
```

3

Deuce & Advantage

Checks for deuce and advantage conditions before displaying normal points. Ensures scoreboard reflects match momentum accurately during competitive play.

```
if (pointsA >= 3 && pointsB >= 3
    && pointsA == pointsB) {
  lcd.print("DEUCE");
}
if (pointsA > pointsB)
  lcd.print("AD Player A");
}
```

Game & Set Win Logic

Program checks for game wins, set wins, and tiebreak wins using official tennis rules—winning by two games or two points. Tiebreak mode entered automatically at 6-6, switching to different scoring system until resolved.

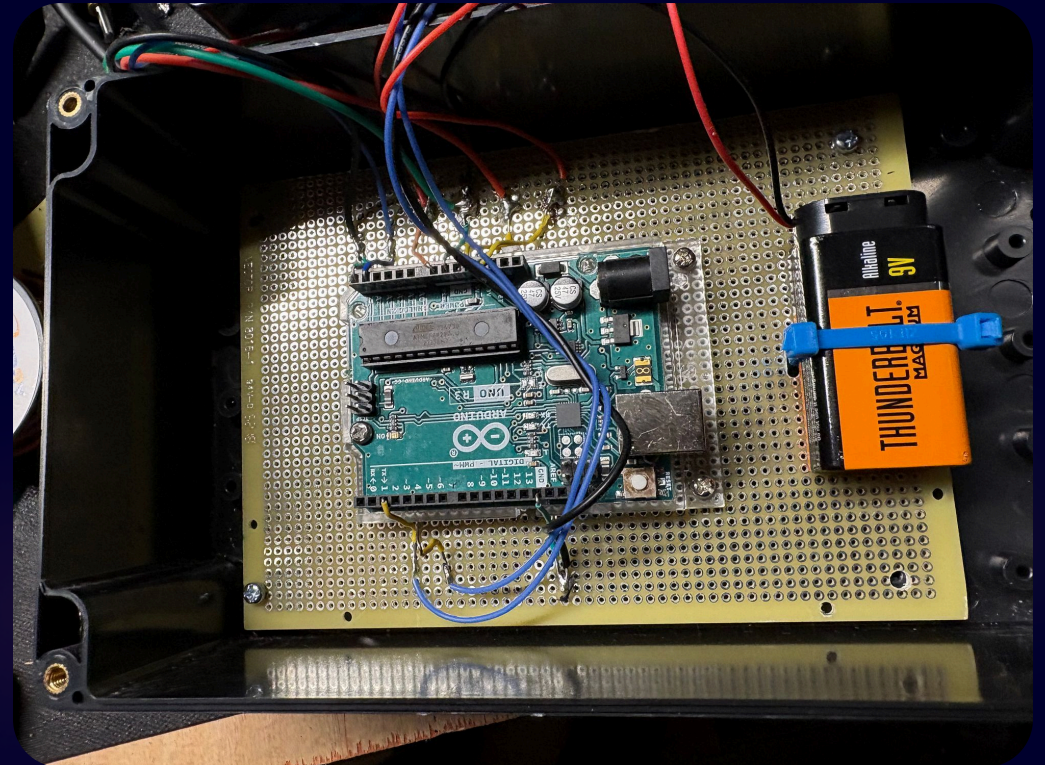
```
if (pointsA >= 4 &&
    pointsA >= pointsB + 2) {
  gamesA++;
  resetPoints();
}
```

Display Management

Single function handles all LCD updates. Separation between logic and display supports future hardware upgrades—easy to replace small LCDs with larger LED panels without rewriting scoring logic.

```
void updateDisplay() {
  lcd.clear();
  if (inTiebreak) {
    lcd.print("TIEBREAK");
  }
}
```

Finished Product



Improvements Over the Prototype

- Display mounted on a ruggedized, drop resistant plastic enclosure. All electronics were moved into a weather-resistant housing.
- All the wired connections have been secured and soldered to a PCB for better reliability.
- On-off switch incorporated to save battery life.
- IR receiver mounted on the box for better line of sight with the IR remote.
- 9V battery used as a power source for the entire setup.

Real-World Testing & Impact

Community Validation

Shared with my high school tennis team and tennis clubs for testing during practice and informal matches. Player feedback validated the need for clear scoring displays on courts lacking infrastructure. This scoreboard is designed around school and club tennis reality—portable, low voltage, and flexible, unlike expensive or permanently installed commercial solutions.

Players appreciated clear visual reference during longer games and deuce situations. Quick score updates reduced verbal confirmation needs and maintained focus. Feedback reinforced designing technology that supports game flow rather than interrupting it.

Future Potential

Strong potential for official matches, tournaments, and practices as display scales and electronics are enclosed for outdoor use. Core design adaptable for other sports requiring simple scorekeeping without permanent installations. Modular architecture allows evolution while preserving original logic.

Impact comes from real use by players who understand competitive tennis pace and pressure. Goal: create a system fitting naturally into school and club sports with potential to become widely usable.

Full Source Code & Next Steps

Access the Code

Complete Arduino source code available in the project repository. Includes all scoring logic, display management, input handling, and tennis rules implementation. Designed for easy modification and hardware upgrades.

[View Repository](#)

I'm sharing the code in the hope that it helps others build, improve, or adapt it for their own communities. I hope it inspires new ideas and helps make tennis a little more enjoyable for everyone!!



Future Development

→ 5G Wireless Capability

Enhanced wireless capabilities such as use of 5GHz WiFi could be used for wireless transmission of the score to a large LED screen for outdoor use.

→ Solar Power

Sustainable energy system for extended outdoor use which would involve solar arrays, a DC energy storage device such as a battery and an inverter.

→ Multi-Court Support

The technology can be scaled for use on multiple courts in tournament environments.

→ Display Scaling

Replace small LCDs with large LED modules or segmented digits. Maintain control logic while increasing readability from across the court.